Pakistan Infrastructure, Logistics & Grid Registry

A highly specialized, technical dossier detailing the foundational transmission, freight vectors, and financial infrastructure driving the country's macroscopic operations.

1. High-Voltage Power Transmission Grid Architecture

The backbone of Pakistan's bulk power transfer, shifting generation loads from southern thermal and nuclear clusters to high-demand northern industrial centers.

Transmission Asset	Technical Specifications	Strategic / Operational Role
Matiari-Lahore HVDC Line	±660 kV DC, 4,000 MW Capacity, ~886 km total circuit length.	Evacuates massive power loads from southern coal and nuclear plants (Thar, KANUPP) directly to northern distribution grids.
CASA-1000 Project	500 kV HVAC/HVDC cross-border linkage.	Designed to import 1,300 MW of clean hydropower from Kyrgyzstan and Tajikistan via Afghanistan into Pakistan's national grid at Nowshera.
Neelum-Jhelum Link	500 kV transmission double-circuit asset.	Evacuates 969 MW of run-of-the-river green energy from the high-altitude Neelum-Jhelum hydro installation directly into the central pool.

2. Deep-Sea Maritime Freight Logistics & Shipping Lanes

Pakistan's trade lifelines connecting the Arabian Sea to global sea lines of communication (SLOC).

Maritime Node / Lane	Technical Parameter / Depth	Throughput Dynamics
Port Muhammad Bin Qasim	13-meter navigable channel draught; contains 12+ deep-water berths.	Handles over 35% of national cargo cargo volume, specializing in bulk industrial inputs, LNG fuel, raw minerals, and heavy machinery.
Karachi Port Trust (KPT)	Natural deep-water harbor, 11.5 to 13-meter draught limits.	Historically the busiest portal; handles diverse container traffic, liquid petroleum raw shipments, and general dry cargo imports.
Gwadar Deep-Sea Port	Design capability up to 14.5 meters; specialized multi-purpose berths.	The linchpin of CPEC maritime connectivity, offering a direct transshipment gateway bypassing the Malacca Strait choke points.

3. Advanced Macroeconomic Registries & Currency Architecture

The technical systems tracking currency valuation, reserve assets, and fiscal minting.

System / Institution	Operational Mandate	Infrastructure Capacity
Pakistan Security Printing Corp (PSPC)	Sole sovereign authority for minting banknotes, secure identity items, passport booklets, and high-security state instruments.	State-of-the-art secure plant in Karachi integrating micro-printing and advanced anti-counterfeiting layers.
SBP FX Reserve Management	Liquidity management across global hard currencies (USD, EUR, RMB) and gold reserves.	Enforces macroeconomic stability, handles sovereign debt debt payments, and defends standard import cover horizons.